



RODEO NOAA

Acoustic real time report

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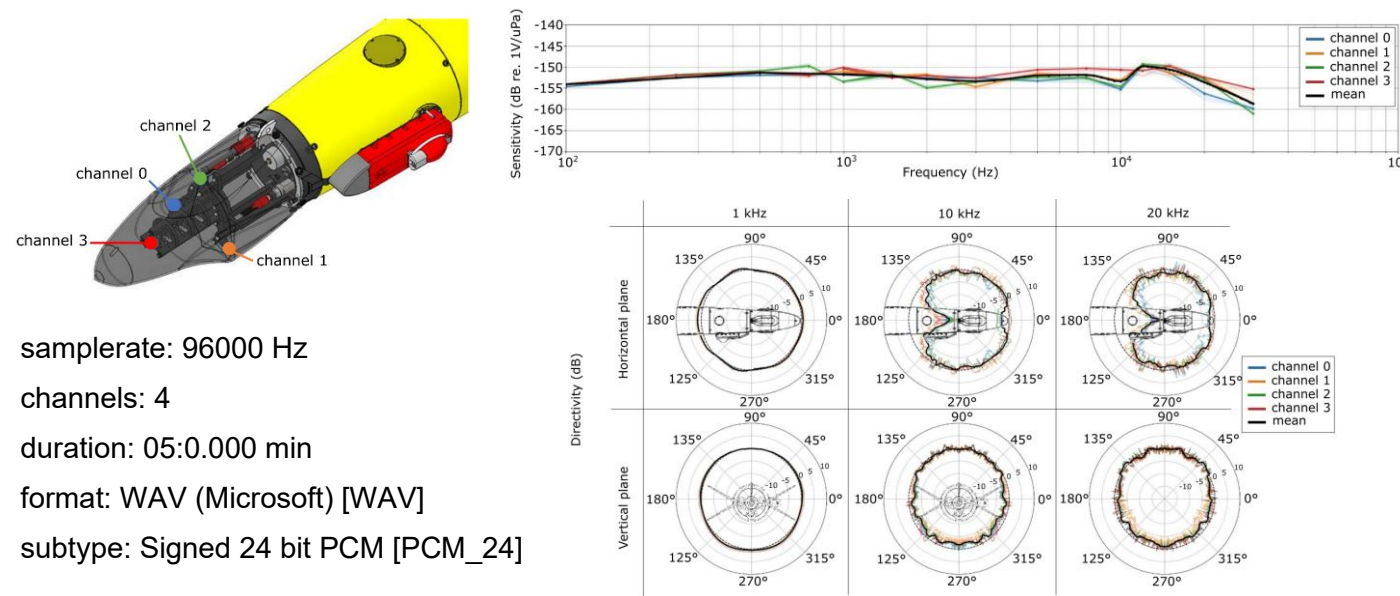
1 INTRODUCTION

1.1 Purpose of the document

The purpose of this document is to provide acoustic observations made during the measurement campaign RODEO February 2026 by the embedded acoustic modules. All the results presented here are obtained only using real time information transmitted by the glider. Raw data are available for download in glider’s storage SSD. Additional resources can be found in the technical report and description of the algorithms of this mission.

1.2 Payload Setup and Configuration

The payload functions both as an acoustic recorder and an onboard processing platform. It allows recording of up to 16 channels and processing of these signals if required. For this mission, data are recorded according to the following settings:



Additionally, three acoustic modules have been added, processing data in parallel for ambient noise monitoring, marine mammals’ detection/classification and support to near real time data analysis during mission operations.

2 SCIENTIFIC ANALYSIS

This report is based solely on the data transmitted in semi-real time by the glider during the RODEO mission. More detailed analyses can be performed by processing the raw data stored on the device.

This semi real time data made it possible to generate a video showing the glider's telemetry, AIS data of surrounding vessels, and a temporal representation of ambient noise data, marine mammal detections, and speed-of-sound profiles. This video is provided as a supplementary material to the written report.

2.1 AMBIA: ambient noise monitoring

AMBIA is a processing module design for the payload AURIS. It is designed to monitor and quantify underwater ambient noise in real time; it continuously analyses data from a hydrophone and computes the sound pressure levels (SPL) across specified frequency bands.

Two types of ambient noise are distinguished:

- **Background noise**, or sea noise, which is a measurement of the ambient noise excluding the instrument's self-noise and identifiable/punctual sounds (source in near field, transient signals...).
- **Ambient noise**, which is a measurement of the ambient noise excluding only the instrument's self-noise.

During the mission we set the glider to measure AMBIENT NOISE over THIRD OCTAVE BANDS providing POWER SPECTRAL DENSITY (dB re. 1µPa²/Hz) every minute. The overview of the ambient noise during the whole mission is available on GLIMPSE and is shown here in Figure 1.

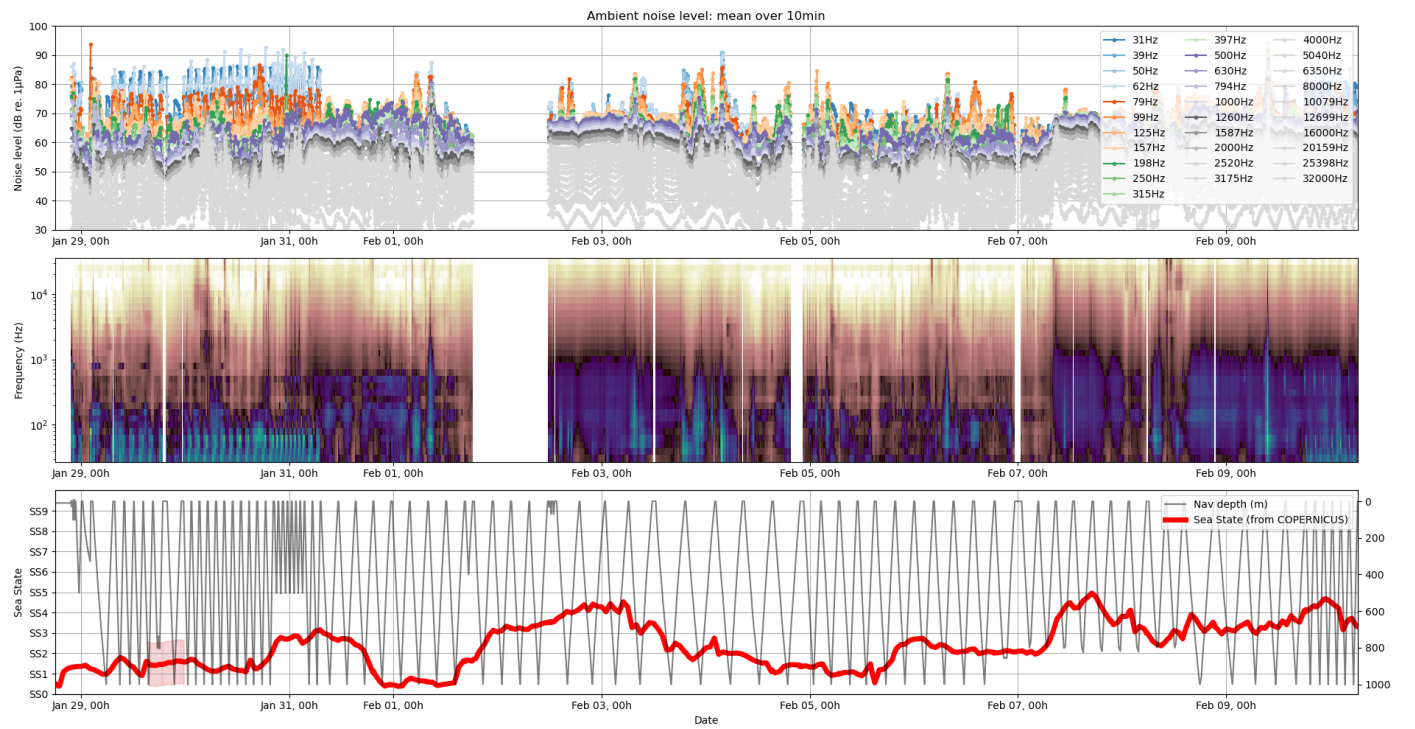


Figure 1: Overview of Ambient Noise provided by Ambia during RODEO mission

From this overview we can explore specific acoustic events. These are developed in the next sections.

Depth–time energy in the octave bands can also be explored in GLIMPSE through the related graphs. Here, we present an example for the octave band centered at 2000 Hz.

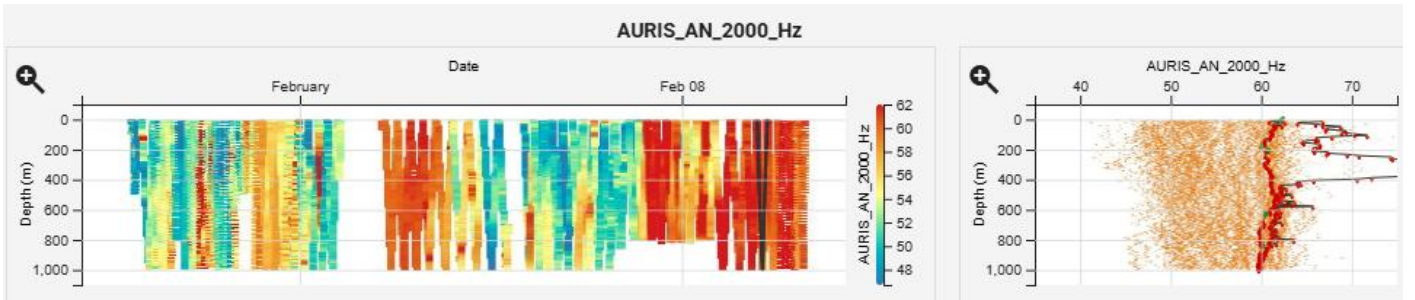


Figure 2: Depth versus time ambient noise at 2000Hz over the whole mission period

2.1.1 Ship radiated noise

When vessels in transit are observed across the entire bandwidth, typically within a 10-minute period, they are identifiable in the ambient noise as transient broadband signatures. Tonal signals from vessels can also be detected at greater distances and over longer time periods, sometimes lasting several hours. We can correlate this acoustic event with AIS data. Few examples are given in Figure 3

Figure 5 : US GOV VESSEL of type cargo vessel exhibited distinct tonal emissions centered in the 800 Hz third-octave band — can be seen from 29 Jan 00:00 to 30 Jan 12:00 UTC

and Figure 4.

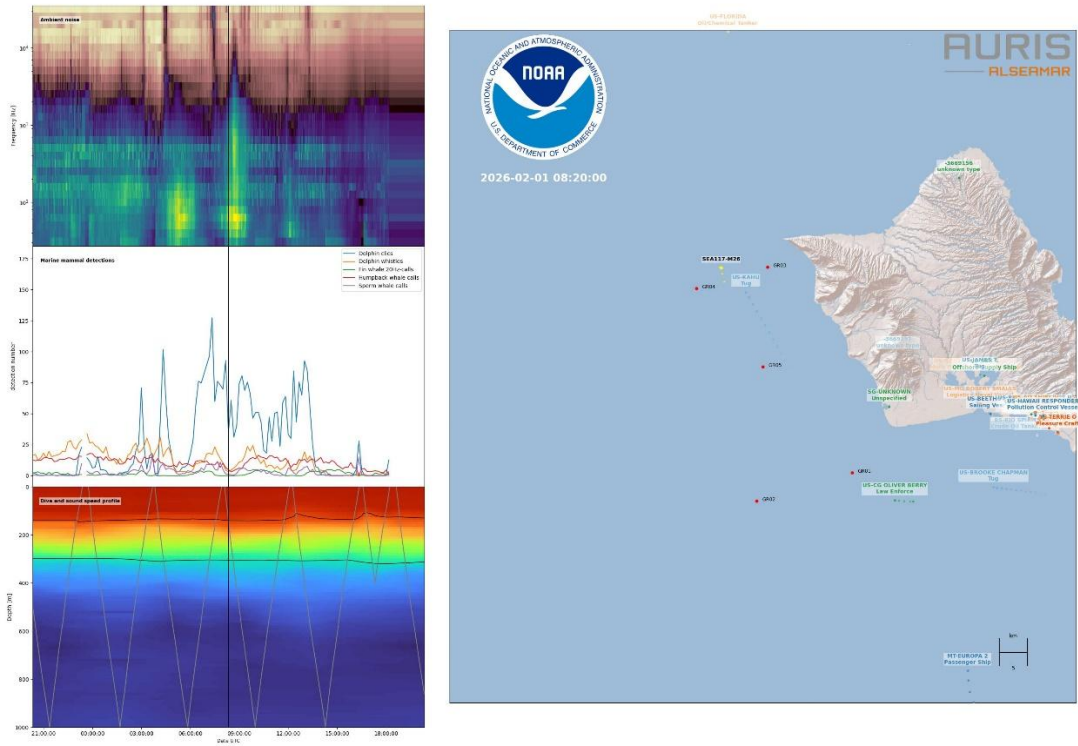


Figure 3: KAHU tug vessel crossing at CPA — 1 Feb, 08:40 UTC

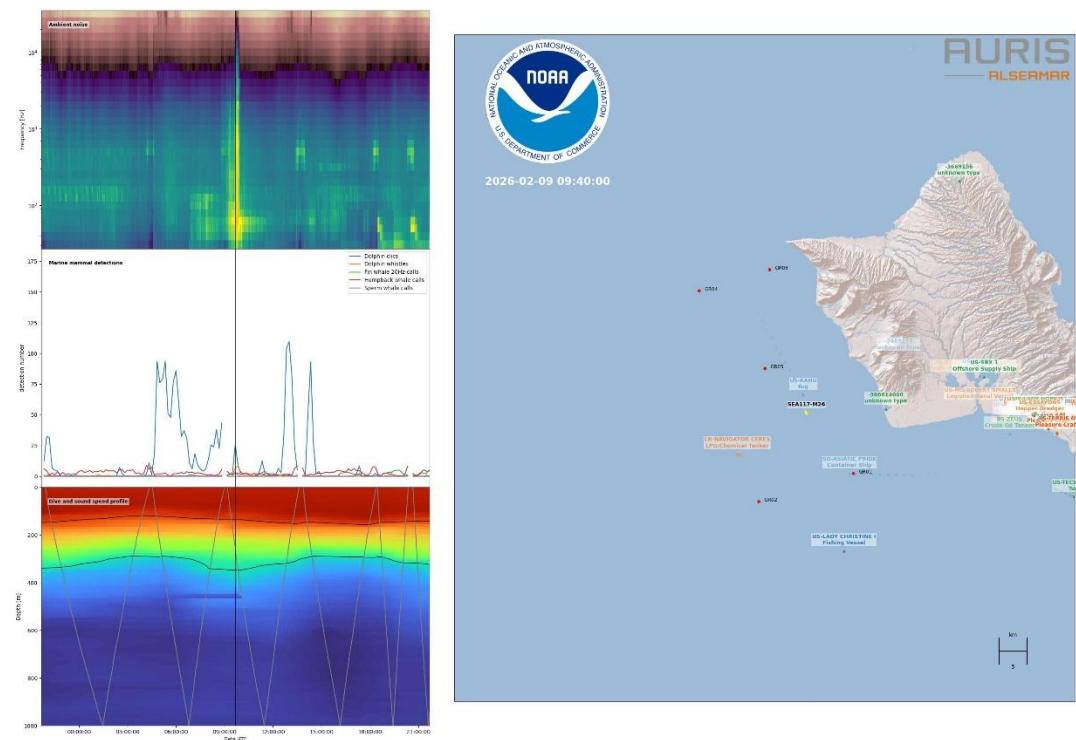


Figure 4: KAHU tug vessel near-collision with glider at 81th surfacing — The glider had been at the surface on the same trajectory 40 minutes prior.

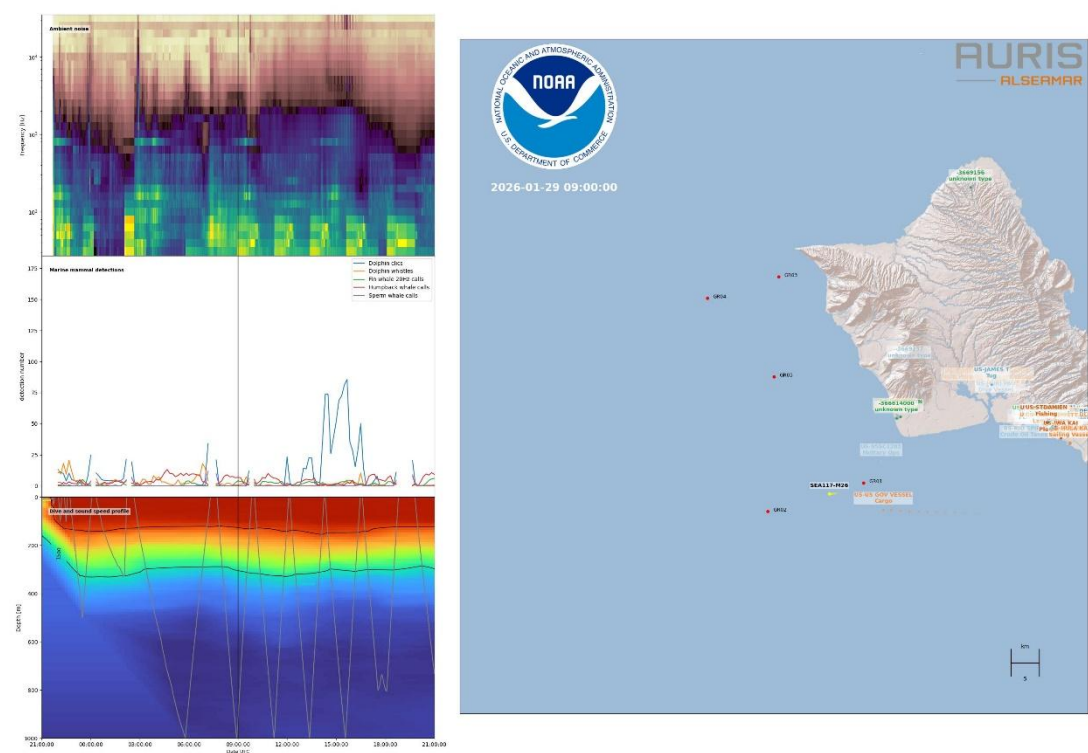


Figure 5 : US GOV VESSEL of type cargo vessel exhibited distinct tonal emissions centered in the 800 Hz third-octave band — can be seen from 29 Jan 00:00 to 30 Jan 12:00 UTC

2.1.2 Flow noise

Gliders are very quiet platforms compared to others equipped with thrusters or connected to a buoy, which generate significant noise (pounding noise, machinery noise, propeller noise, flow noise...). Even though gliders may be considered silent platforms, they are still affected by self-noise, such as flow noise.

On the SeaExplorer, flow noise is reduced using an acoustically transparent nose that redirects the flow around the glider body rather than over the sensor. However, at high speeds (i.e., glider vertical speeds > 20 cm/s), flow noise can still be observed below 100 Hz.

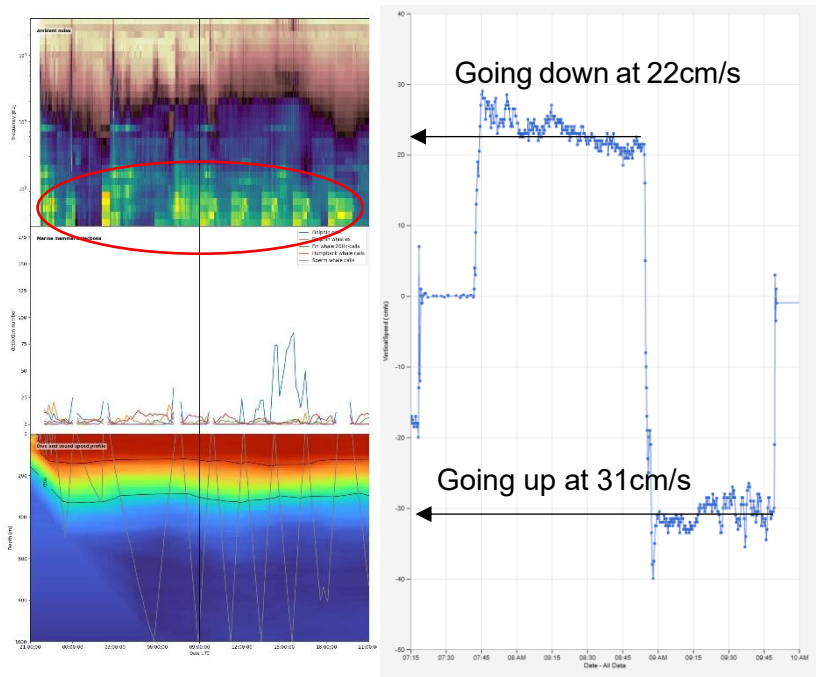


Figure 6: Glider-observed flow noise during high speed. Asymmetric speed with 22cm/s during going down showing very limited flow noise and going up at 31cm/s showing clear flow noise under 100Hz.

Please note that self-noise from actuators is effectively rejected in the embedded ambient noise calculation, and navigation modes do not affect the Ambia estimator; only flow noise has an impact.

Overall, during the RODEO mission, flow noise was observable during high-speed vertical flights at the start of the mission (29–30 January) and again near the end, when approaching the recovery point on 9–10 February. Other periods were highly suitable for low-noise measurements, even in the lower third-octave band at 32 Hz, with the glider operating at medium to low speeds of approximately 10–20 cm/s.

2.1.3 Relation with sea state/marine traffic

Ambient noise is primarily influenced by Sea State (SS), particularly above 500 Hz, as illustrated by Wenz curves relating power spectral levels to sea conditions. Statistics of ambient noise spectrum is also provided on GLIMPSE and superimposed on Wenz theoretical curves

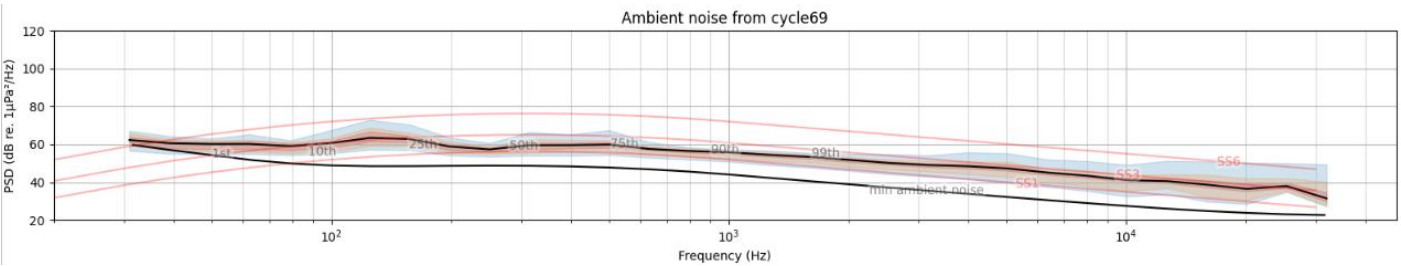


Figure 7: Distribution of ambient noise power spectral levels during Yo69. The black curve represents the median, the blue area indicates the 1st–99th percentiles, the orange area corresponds to the 10th–90th percentiles, and the red area denotes the 25th–75th percentiles.

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Marine traffic primarily affects the lower spectral bands, particularly below 500 Hz. Hawaiï area is associated with very low marine traffic, allowing measurements to approach the theoretical minimum ambient noise for the prevailing sea state most of the time (provided that flow noise is absent).

Variations in sea state were well captured over the mission period. In particular, sea states of SS4 and above, resulting from high wind speeds, occurred during three periods and can be identified by a marked increase in ambient noise, as shown in Figure 8.

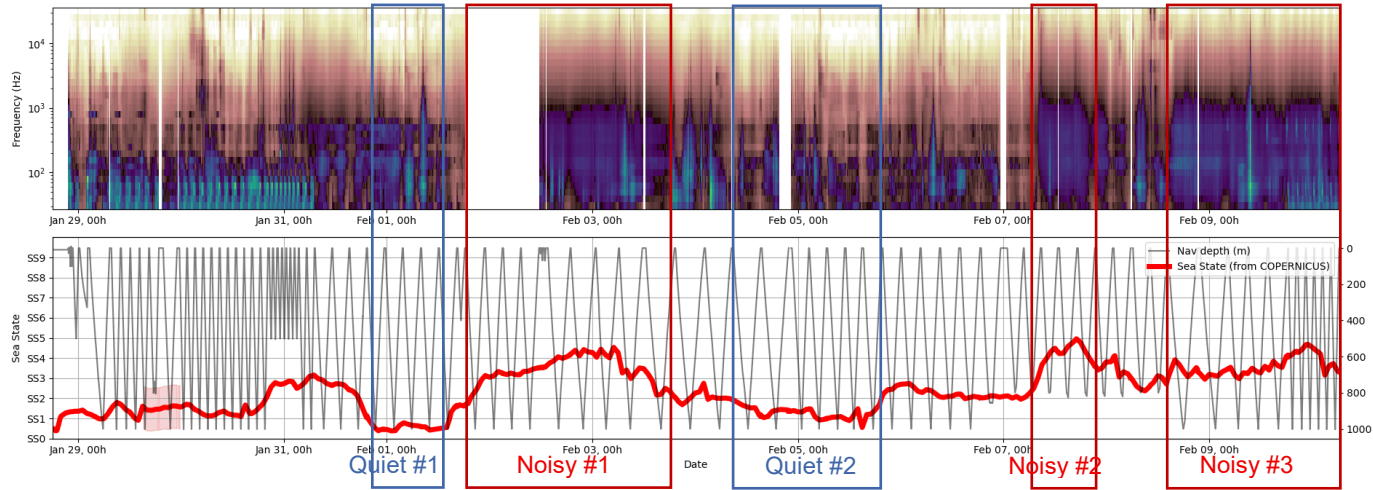


Figure 8 : Relation between power spectral levels and wind speed

The quietest period should have occurred on 1 February, with sea state below SS1. This period, marked with a blue box in Figure 8, corresponds to low background noise, but several ship signatures and a pronounced biological chorus—attributed to humpback whales—are present (see dedicated section for more details). A second period of low background noise occurred the 5th February, this time less punctual sources are present showing lower noise level than 1 February. This period may be of particular interest for those seeking to study low ambient noise conditions.

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2.2 DELPHIA: marine mammals monitoring

DELPHIA is an embedded software module for the AURIS platform that automatically detects marine mammals sounds in real time from hydrophone data. It continuously processes incoming acoustic signals, classifies detections based on species group and signal characteristics, and logs the results. Detection data are transmitted to the GLIMPSE platform at the next surfacing event for visualization and analysis.

Detected Species & Vocalization Types:

- Fin Whales: 20-Hz pulses in ultra-low frequencies (ULF)
- Humpback Whales: calls in low to mid frequencies (LF – MF)
- Sperm Whales: clicks in mid frequencies (MF)
- Dolphins: whistles in mid-high frequencies (MF – HF)
- Dolphins: clicks in high frequencies (HF)

The module is based on two distinct algorithms.

- 1- Detection is performed on the spectrogram, identifying statistical anomalies in the signal's time-frequency representation. It acts as a general anomaly detector. Then, all these anomalies are labelled in terms of signal type: frequency modulation, transient, broad band, pulse or continuous wave.
- 2- Anomalies are given to the DELPHIA classifier. The classifier is based on multiple Gaussian Mixture Models (GMMs), one per class to be identified.

Possible labels are:

- DW (dolphin whistles)
- DC (dolphin clicks)
- FW (fin whale 20-Hz calls)
- HW (humpback whale calls)
- SW (sperm whale clicks)
- Unknown (if no class attributed)

2.2.1 Mission overview

During the RODEO mission, the glider detected, with a confidence index of 100%, four of the five target marine mammal species: SW, DC, DW, and HBW. FW may have been detected; however, these detections were not confirmed upon spectrogram inspection.

In fact, during each dive, a 30-second low-resolution spectrogram sample—where the maximum number of detections is recorded—is sent along with the detection file. These spectrograms can be rapidly reviewed by an analyst directly in GLIMPSE to corroborate the algorithm's classification. We present some of these spectrograms later in this section.

Please note that this inspection could have been supported by the use of a high-resolution spectrogram request module. This module was designed to generate a spectrogram from the recordings according to user-specified parameters. It is therefore a module that operates on an interrupt triggered by the reception of a user command. Unfortunately, this module did not function properly during the RODEO mission, likely due to either a failed module update just before the mission or an incompatibility between the navigation software and the payload. In either case, updates performed a few days prior to the mission, as well as final pre-launch checks, should be improved to detect such errors in advance. However, it did not affect other aspects of the mission, since the modules operate largely independently thanks to the ROS (robotic operating system) software architecture.

Number of detections per seconds over the whole mission is presented on Figure 9. Obviously, the metric is related to natural behaviour and the number of vocalizing individuals. For example, delphinids produce more signals per second when echolocating than sperm whales or humpback whales do when acoustically active.

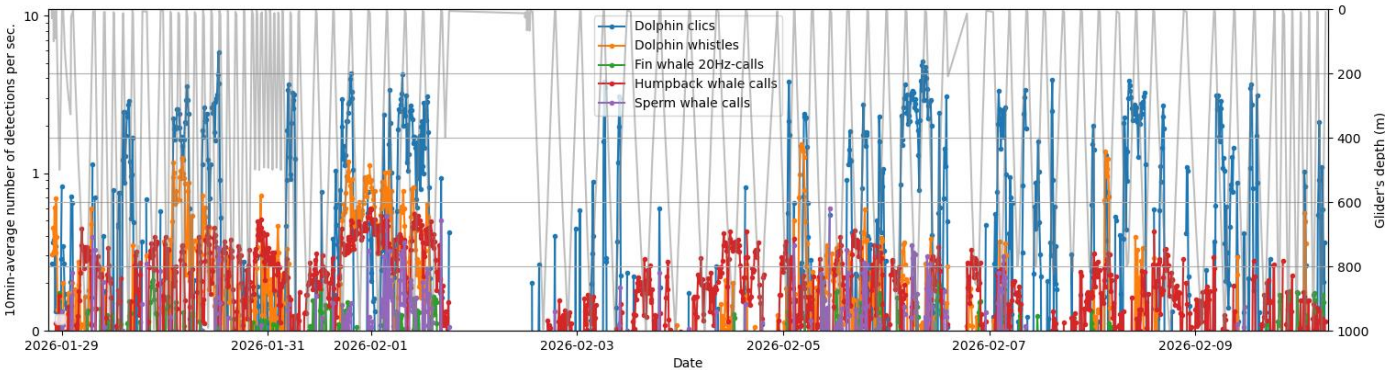


Figure 9 : Number of detections per seconds — mean over 10 minutes.

Since the glider data are depth–time related, it is often more appropriate to visualize them as a depth versus time scatter plot, as illustrated in Figure 10. Please note that the color threshold is set by class at the 90th percentile and therefore does not correspond to the same detection rate across classes.

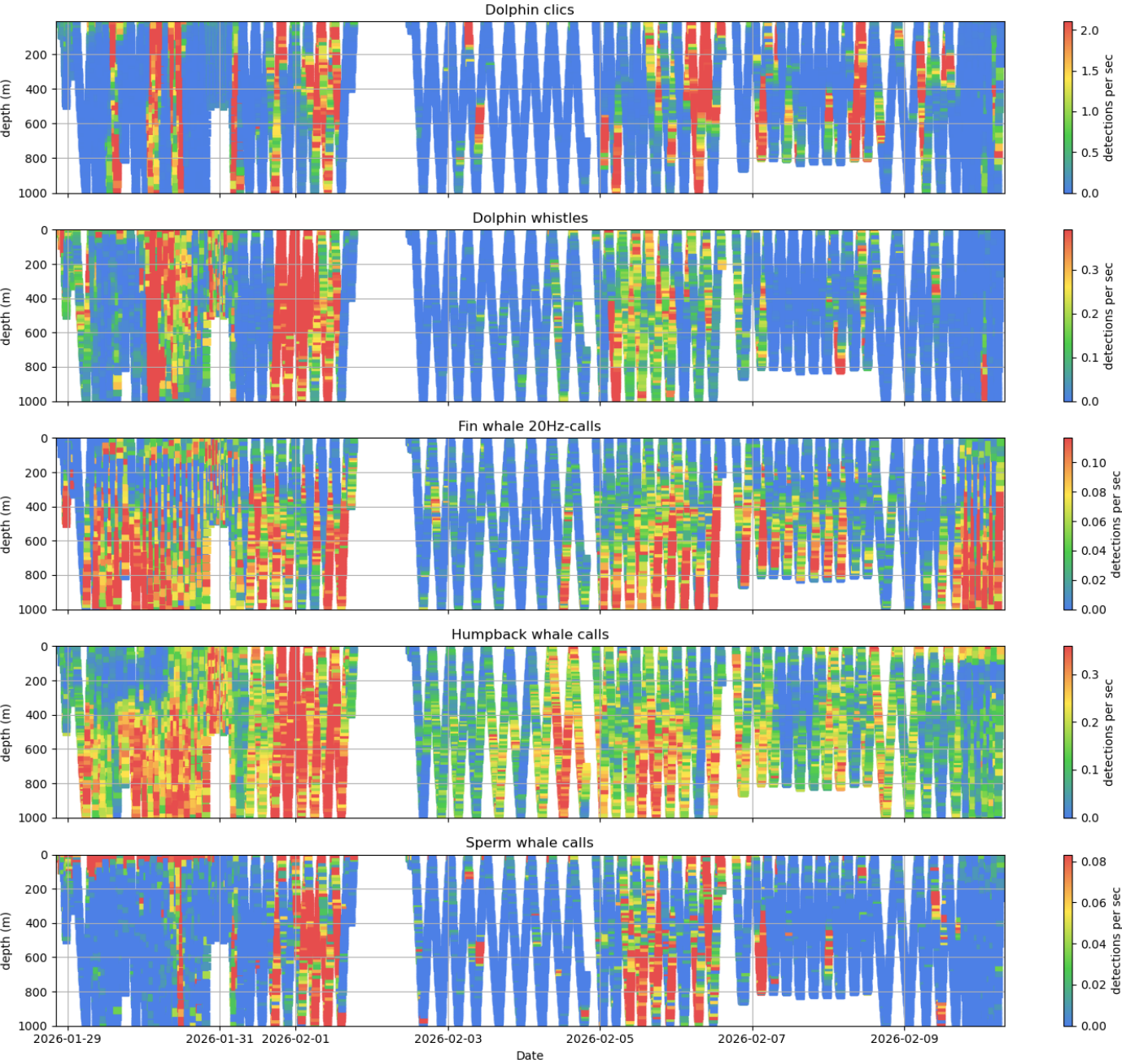


Figure 10 : Detections profile per class

In this overall view, high biological activity is evident, particularly on 1 February, when sperm whales, humpback whales, and delphinids appear to be acoustically very active. This occurs concomitantly with a low sea state (might be related biological behaviour and/or better SNR increase detections).

2.2.2 Examples of spectrogram of targeted species

Among the collection of low-resolution spectrograms transmitted by the glider, the following examples best represent each class:

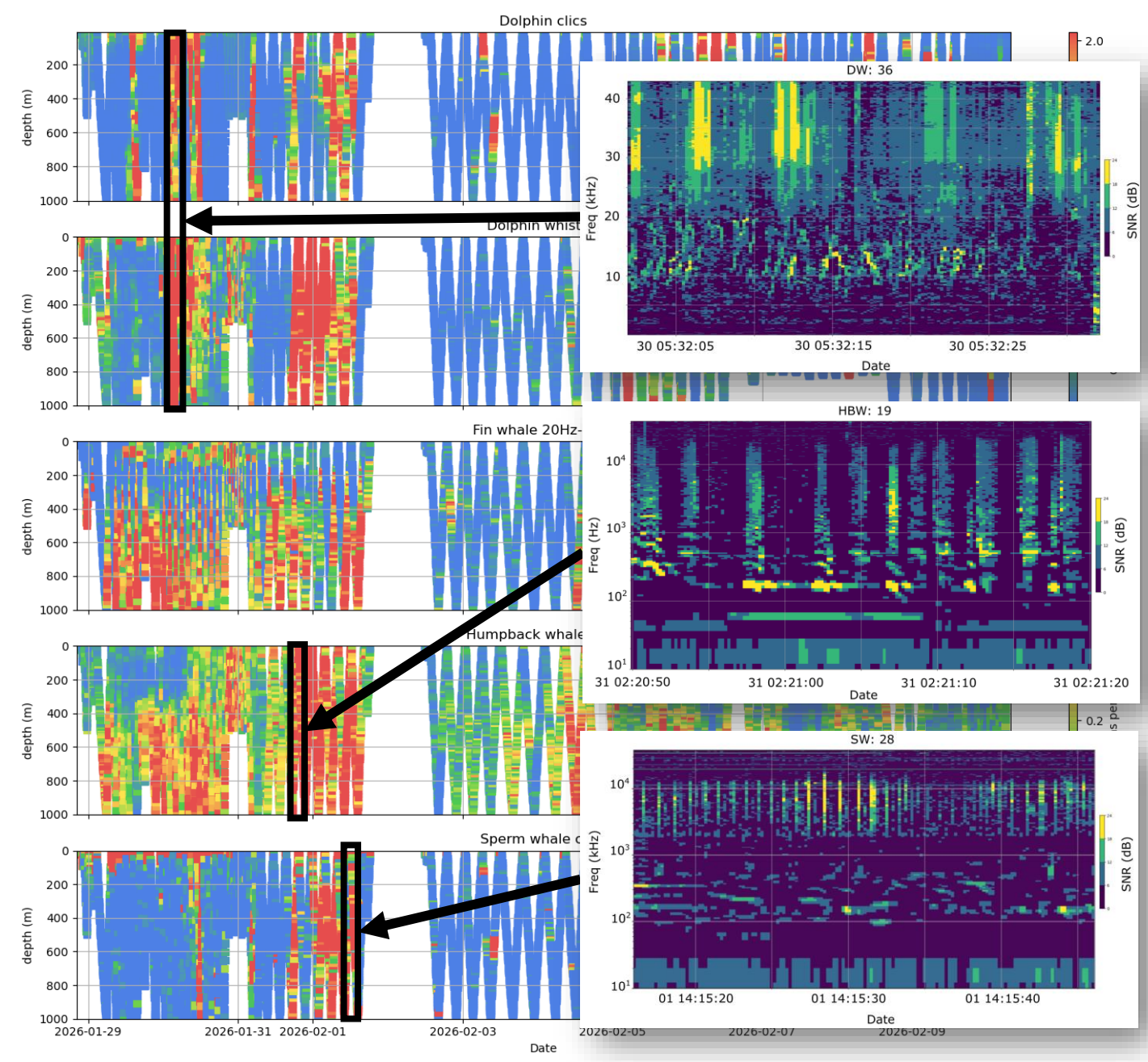


Figure 11 : Overview of all detection. Low resolution spectrogram are provided to illustrate Delphinids clicks/whistles, humpback whales calls and sperm whales echo localization clicks.

We do have some anomalies around 20Hz classified as 20Hz-Fin whale calls but spectrograms lack of resolution to clearly confirm the signature. Example of the 31 Jan where 14 20Hz-calls have been reported:

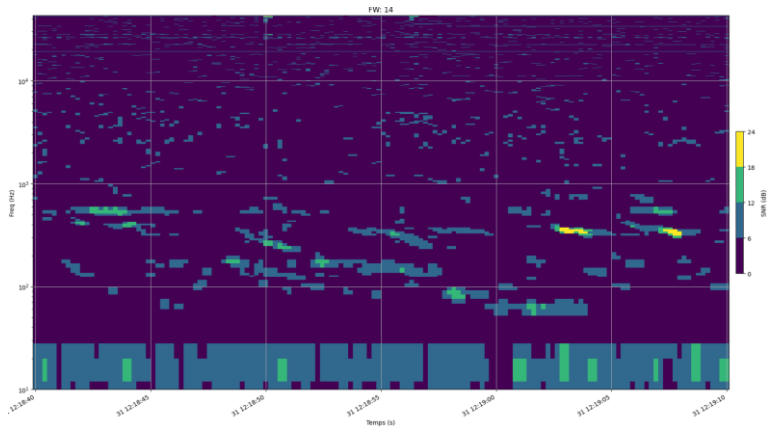


Figure 12 : examples of classified fin whale’s 20Hz-calls — hard to confirm detection

2.2.3 Depth relation of the detection rate

Here we highlight the dependence of detections on glider depth. Using the sound speed profile continuously collected by the glider, long-range acoustic propagation effects can be identified. This is particularly evident at the start of the mission, when the number of humpback whale detections increases significantly below 400m depth.

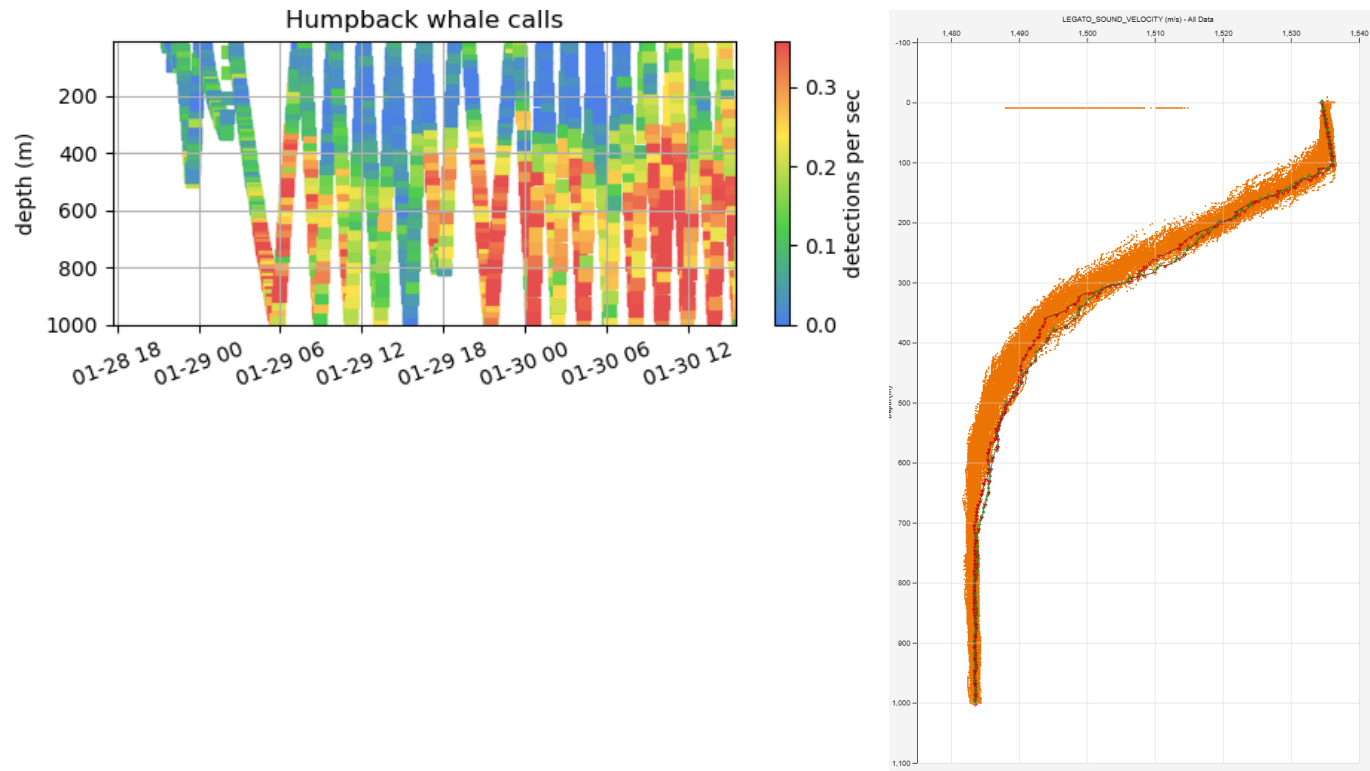


Figure 13: humpback whale detections with respect to sound speed profile

This may indicate long-range detection, where sound propagates below the thermocline. When marine mammals are closer, this effect is no longer apparent, and detections occur regardless of the glider’s depth.

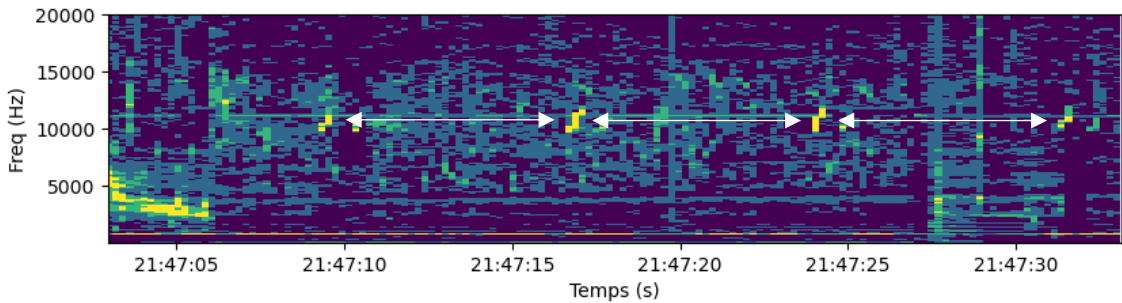
2.2.4 False alarms

From the data, false alarms can be identified by examining unusual relationships between detections and unrelated variables. For example, we observe:

- Increased fin whale detections when the glider is moving rapidly (flow noise generates numerous low-frequency acoustic anomalies, including in the 20 Hz band).
- Increased detections of delphinid whistles when humpback whales are very close (call harmonics produce frequency modulations up to ~10 kHz that can resemble dolphin whistles).
- Increased click detections (sperm whales and delphinids) when the glider is near the surface (cause not yet identified).
- Increased sperm whale click detections when a vessel passes nearby (propeller cavitation impulses can share similar time–frequency signatures).

2.2.5 Other acoustic activities

Since we retrieve the full background noise context, rather than only detected signals, some low-resolution spectrograms transmitted by the glider may include may include unattended acoustic signatures. For example, active sonar emissions can be observed among delphinid whistles. The high signal-to-noise ratio and constant repetition rate clearly indicate that these signals are not biological. It is unclear whether the classifier identified these acoustic anomalies as delphinid whistles or simply as unknown signals. It is likely that they were counted as whistles.



Another example is this isolated acoustic signature that was attributed to delphinid whistles. However, the harmonic features suggest that they may have been misclassified and could originate from another species not recognized by the classifier. This has been recorded the 10 February 02:30 UTC just before recovery. A similar signature, exhibiting additional harmonic components, was observed on 30 January at 08:54 UTC.

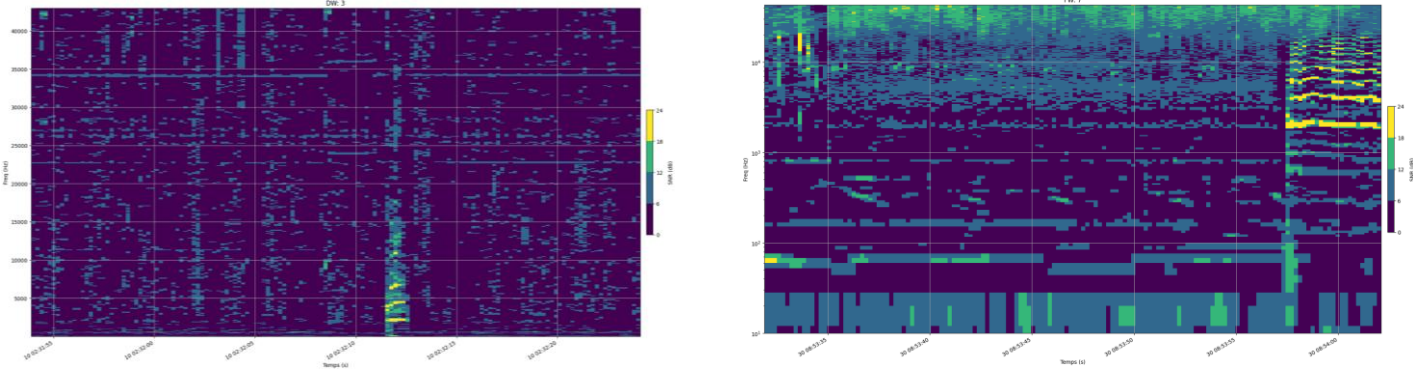


Figure 14 : Low freq harmonic frequency modulated signal

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3 CONCLUSION

Real-time algorithms completed the RODEO mission. The acoustic anomaly detector, coupled with the DELPHIA classifier, captured signatures from sperm whales, humpback whales, and delphinid clicks and whistles. Fin whale detections may have occurred but could not be confirmed without inspecting the raw data.

These detections are useful for near-real-time analysis and provide sufficiently accurate information to identify acoustic activity hotspots for the different species. The low-resolution spectrograms automatically transmitted to support detection validation proved very useful and sufficient to confirm detections with a confidence index of 100% (except for fin whale 20 Hz calls).

Ambient noise monitoring also provided valuable and verified acoustic features of the area, which can support near-real-time decision-making during mission operations. Flow noise was observed during high-speed flights but was greatly reduced or absent at slower speeds (vertical speed < 20 cm/s).

Even though part of the raw data were lost—likely due to an SSD connection failure—the full mission has been “fortunately” completed thanks to the real-time algorithms. Also, some real-time features, such as on-demand high-resolution spectrograms and azimuth computation, were unavailable. Both issues were caused by software misconfigurations on the glider, either due to incompatibilities between navigation and payload software versions or incorrect configuration files. In either case, updates performed a few days prior to the mission, as well as final pre-launch checks, should be improved to detect such errors in advance and to ensure full functionality.